6 The table shows the half-life of various radioactive nuclides.

nuclide	$^{197}_{84}Po$	$^{220}_{86}Rn$	$^{222}_{88}Ra$	$^{234}_{91}Pa$
half-life / s	60	52	38	70

The variation of the number of nuclei with time for one of these radioactive nuclides is shown in Fig. 6.1.

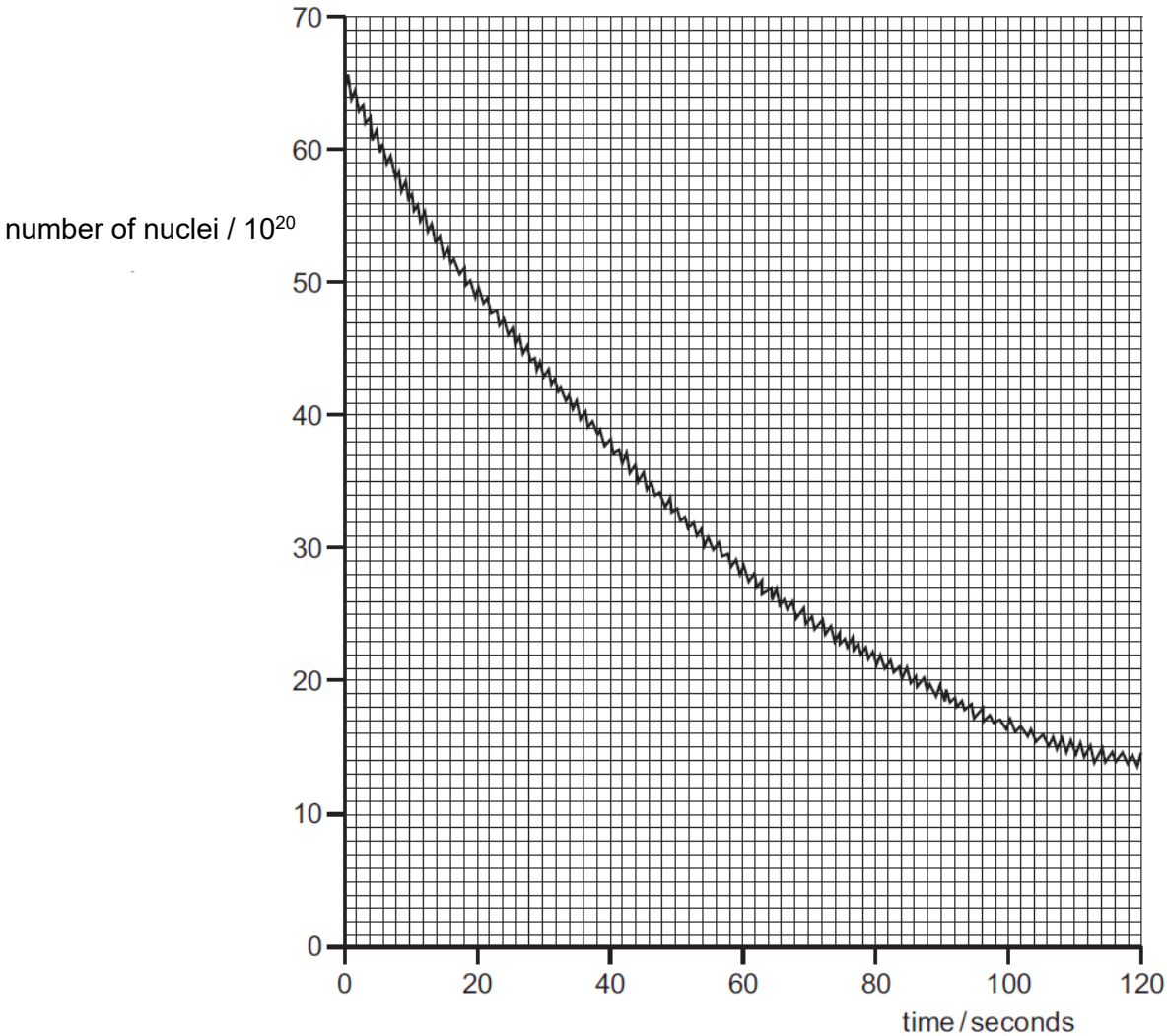

Fig. 6.1

(a) **(i)** Using Fig. 6.1, determine which is the decaying radioactive nuclide. Show your working clearly.

decaying radioactive nuclide = …… [3]

(ii) Hence, calculate the initial activity of the radioactive nuclide.

initial activity = …… Bq [2]

The decaying radioactive nuclide emits an alpha particle as it decays into a stable daughter nuclide Y.

(b) By plotting 3 points on Fig. 6.1, sketch a graph of the variation of number of daughter nuclide Y with time. Show your working clearly.

[2]

(c) Write a complete nuclear equation for the decay process.

[1]

(d) Given that linear momentum is conserved in this decay, show that the ratio $\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus Y}} = \frac{1}{4}A - 1$ where A is the nucleon number of the decaying radioactive nuclide.

[2]